
GOOGLE PLAY STORE DATA **ANALYSIS USING PYTHON**

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Course : B.Com Computer Application

Tools used : Python, pandas, NumPy, Matplotlib, seaborn, Google CoLab

Project Type : Data Analytics Project

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- **INTRODUCTION**

GOOGLE PLAY STORE IS A POPULAR PLATFORM THAT PROVIDES A WIDE VARIETY OF MOBILE APPLICATIONS FOR ANDROID USERS. THESE APPLICATIONS BELONG TO DIFFERENT CATEGORIES SUCH AS COMMUNICATION, ENTERTAINMENT, EDUCATION, GAMES, BUSINESS, PHOTOGRAPHY, AND PRODUCTIVITY.

THIS PROJECT FOCUSES ON ANALYZING GOOGLE PLAY STORE DATA USING PYTHON. THE DATASET CONTAINS INFORMATION ABOUT APPLICATIONS, INCLUDING RATINGS, REVIEWS, NUMBER OF INSTALLATIONS, APP SIZE, PRICE, CATEGORY, AND OTHER DETAILS.

THE PURPOSE OF THIS PROJECT IS TO EXPLORE THE DATASET, CLEAN THE AVAILABLE DATA, IDENTIFY PATTERNS, AND UNDERSTAND FACTORS RELATED TO APP POPULARITY AND USER ENGAGEMENT. PYTHON LIBRARIES SUCH AS PANDAS, NUMPY, MATPLOTLIB, AND SEABORN ARE USED FOR DATA CLEANING, ANALYSIS, AND VISUALIZATION.

THE ANALYSIS PROVIDES USEFUL INSIGHTS INTO APP INSTALLATIONS, REVIEWS, RATINGS, PRICING, AND STORAGE REQUIREMENTS. THESE INSIGHTS CAN HELP UNDERSTAND USER PREFERENCES AND APP PERFORMANCE.

- PROJECT OBJECTIVE

The main objective of this project is to analyze Google Play Store data using Python and identify useful patterns and insights from the dataset. The project focuses on understanding app popularity, user engagement, ratings, installations, pricing, and storage requirements.

The key objectives of the project are:

- * To clean and prepare the Google Play Store dataset for analysis.*
- * To analyze app categories and their characteristics.*
- * To identify the most installed applications.*
- * To analyze applications with the highest number of reviews.*
- * To examine app ratings and user feedback.*
- * To analyze free and paid applications.*
- * To identify applications with large file sizes.*
- * To create visualizations for better understanding of the data.*

** To derive useful insights that can help understand app popularity and user engagement.*

3. DATASET DESCRIPTION

THE DATASET USED IN THIS PROJECT IS GOOGLE PLAY STORE APP DATA. IT CONTAINS INFORMATION ABOUT ANDROID APPLICATIONS AVAILABLE ON THE GOOGLE PLAY STORE. THE DATASET PROVIDES DETAILS ABOUT DIFFERENT ASPECTS OF THE APPLICATIONS, SUCH AS CATEGORY, RATING, REVIEWS, INSTALLATIONS, SIZE, PRICE, CONTENT RATING, AND LAST UPDATED DATE.

THE DATASET CONTAINS THE FOLLOWING IMPORTANT COLUMNS:

- * APP: NAME OF THE APPLICATION.
- * CATEGORY: CATEGORY TO WHICH THE APPLICATION BELONGS.
- * RATING: AVERAGE USER RATING OF THE APPLICATION.
- * REVIEWS: NUMBER OF USER REVIEWS RECEIVED BY THE APPLICATION.
- * SIZE: SIZE OF THE APPLICATION.
- * INSTALLS: NUMBER OF INSTALLATIONS OF THE APPLICATION.
- * TYPE: INDICATES WHETHER THE APPLICATION IS FREE OR PAID.

- * PRICE: PRICE OF THE APPLICATION.
- * CONTENT RATING: TARGET AUDIENCE RATING OF THE APPLICATION.
- * GENRES: GENRE OR TYPE OF APPLICATION.
- * LAST UPDATED: DATE WHEN THE APPLICATION WAS LAST UPDATED.
- * CURRENT VER: CURRENT VERSION OF THE APPLICATION.
- * ANDROID VER: MINIMUM ANDROID VERSION REQUIRED TO RUN THE APPLICATION.

THIS DATASET IS USED TO ANALYZE APP POPULARITY, USER ENGAGEMENT, RATINGS, PRICING, AND OTHER IMPORTANT CHARACTERISTICS OF APPLICATIONS.

4. TOOLS & TECHNOLOGIES

THE FOLLOWING TOOLS AND TECHNOLOGIES WERE USED TO COMPLETE THIS PROJECT:

PYTHON: USED AS THE MAIN PROGRAMMING LANGUAGE FOR DATA CLEANING, ANALYSIS, AND PROCESSING.

PANDAS: USED FOR LOADING, CLEANING, MANIPULATING, AND ANALYZING THE DATASET.

NUMPY: USED FOR NUMERICAL OPERATIONS AND DATA PROCESSING.

MATPLOTLIB: USED TO CREATE CHARTS AND GRAPHS FOR DATA VISUALIZATION.

SEABORN: USED TO CREATE STATISTICAL AND ATTRACTIVE DATA VISUALIZATIONS.

GOOGLE COLAB: USED AS THE DEVELOPMENT ENVIRONMENT TO WRITE AND EXECUTE PYTHON CODE.

5. DATA CLEANING

DATA CLEANING IS AN IMPORTANT STEP IN DATA ANALYSIS. IT HELPS TO IMPROVE THE QUALITY AND ACCURACY OF THE DATASET BEFORE PERFORMING ANALYSIS. IN THIS PROJECT, THE GOOGLE PLAY STORE DATASET WAS CLEANED AND PREPARED USING PYTHON AND PANDAS.

THE FOLLOWING DATA CLEANING STEPS WERE PERFORMED:

1. HANDLING MISSING VALUES:

MISSING VALUES IN IMPORTANT COLUMNS SUCH AS RATING WERE IDENTIFIED AND HANDLED APPROPRIATELY.

2. CLEANING THE INSTALLS COLUMN:

THE INSTALLS COLUMN CONTAINED VALUES SUCH AS "10,000+", "500,000+", AND "1,000,000+". THE PLUS SIGN (+) AND COMMAS WERE REMOVED, AND THE VALUES WERE CONVERTED INTO NUMERICAL FORMAT FOR ANALYSIS.

3. CLEANING THE PRICE COLUMN:

THE PRICE INFORMATION WAS CHECKED AND PREPARED SO THAT FREE AND PAID APPLICATIONS COULD BE ANALYZED.

4. CHECKING DATA TYPES:

THE DATA TYPES OF IMPORTANT COLUMNS WERE CHECKED AND CONVERTED WHERE NECESSARY SO THAT NUMERICAL ANALYSIS COULD BE PERFORMED CORRECTLY.

5. CHECKING OR HANDLING INVALID DATA:

DUPLICATE OR INCONSISTENT RECORDS AND INVALID VALUES WERE CHECKED TO IMPROVE THE RELIABILITY OF THE ANALYSIS.

AFTER DATA CLEANING, THE DATASET WAS READY FOR EXPLORATORY DATA ANALYSIS AND VISUALIZATION.

6. EXPLORATORY DATA ANALYSIS

EXPLORATORY DATA ANALYSIS (EDA) IS USED TO UNDERSTAND THE MAIN CHARACTERISTICS AND PATTERNS PRESENT IN THE DATASET. IN THIS PROJECT, THE GOOGLE PLAY STORE DATASET WAS EXPLORED TO UNDERSTAND APP POPULARITY, USER ENGAGEMENT, RATINGS, INSTALLATIONS, REVIEWS, AND APP SIZE.

THE FOLLOWING ANALYSES WERE PERFORMED:

1. APP SIZE ANALYSIS:

THE APPLICATIONS WERE ANALYZED BASED ON THEIR SIZE TO IDENTIFY APPS WITH LARGE FILE SIZES.

2. INSTALLATION ANALYSIS:

THE NUMBER OF INSTALLATIONS WAS ANALYZED TO IDENTIFY THE MOST INSTALLED APPLICATIONS AND UNDERSTAND APP POPULARITY.

3. REVIEW ANALYSIS:

THE NUMBER OF REVIEWS WAS ANALYZED TO IDENTIFY APPLICATIONS WITH HIGH USER ENGAGEMENT.

4. RATING ANALYSIS:

APP RATINGS WERE ANALYZED TO IDENTIFY HIGHLY RATED APPLICATIONS AND UNDERSTAND USER SATISFACTION.

5. FREE AND PAID APP ANALYSIS:

APPLICATIONS WERE COMPARED BASED ON THEIR TYPE AND PRICING TO UNDERSTAND THE DIFFERENCE BETWEEN FREE AND PAID APPS.

EDA RESULTS

THE ANALYSIS SHOWS THAT SOME APPLICATIONS HAVE VERY HIGH INSTALLATION NUMBERS AND REVIEWS. FREE APPLICATIONS GENERALLY HAVE HIGHER USER ADOPTION, WHILE POPULAR SOCIAL AND COMMUNICATION APPLICATIONS RECEIVE A LARGE NUMBER OF REVIEWS. THE ANALYSIS ALSO IDENTIFIED HIGHLY RATED APPLICATIONS AND APPLICATIONS WITH LARGE FILE SIZES.

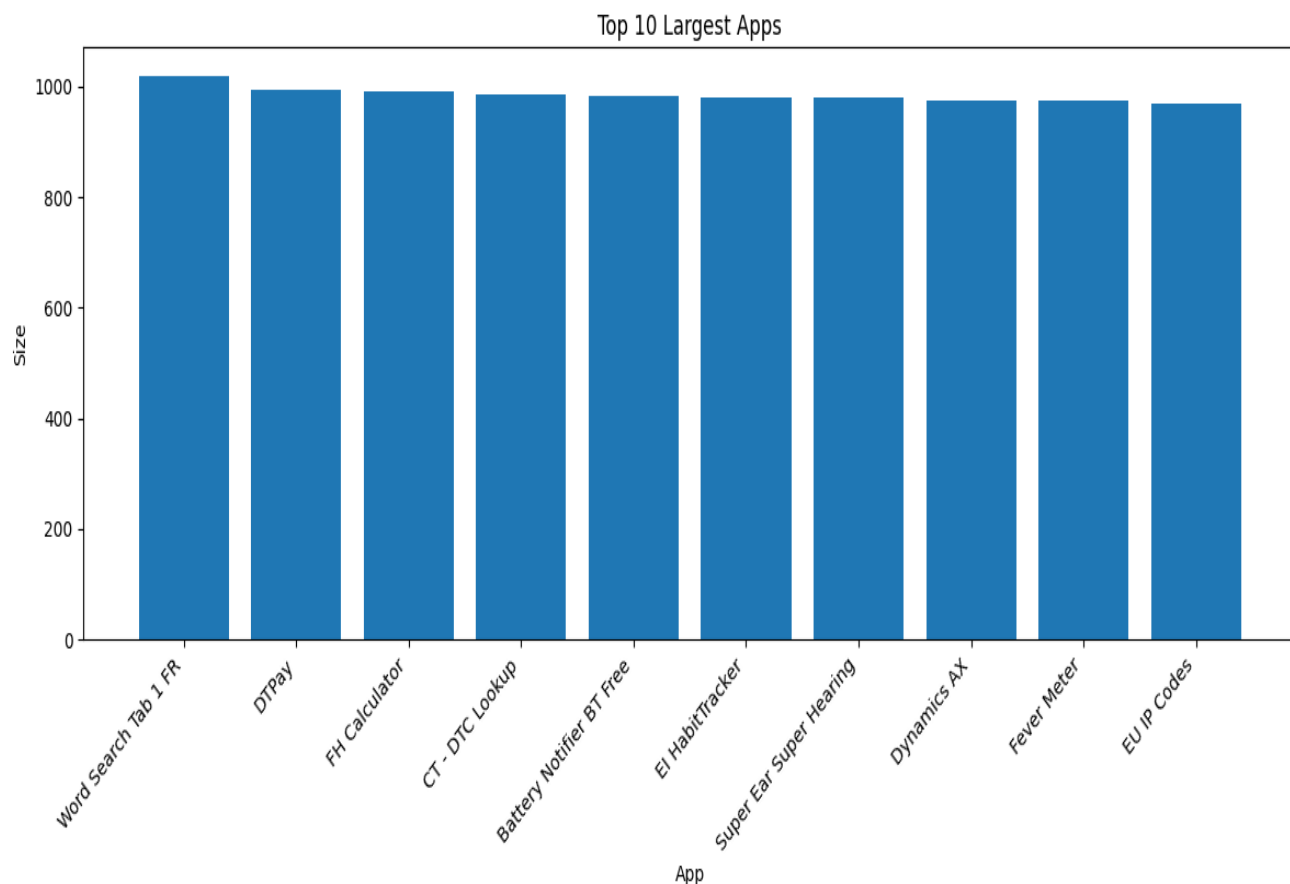
7. DATA VISUALIZATION

DATA VISUALIZATION IS USED TO PRESENT THE RESULTS OF DATA ANALYSIS IN THE FORM OF CHARTS AND GRAPHS. IN THIS PROJECT, PYTHON LIBRARIES SUCH AS MATPLOTLIB AND SEABORN WERE USED TO CREATE VISUALIZATIONS AND UNDERSTAND THE GOOGLE PLAY STORE DATA MORE EASILY.

THE FOLLOWING VISUALIZATIONS WERE CREATED:

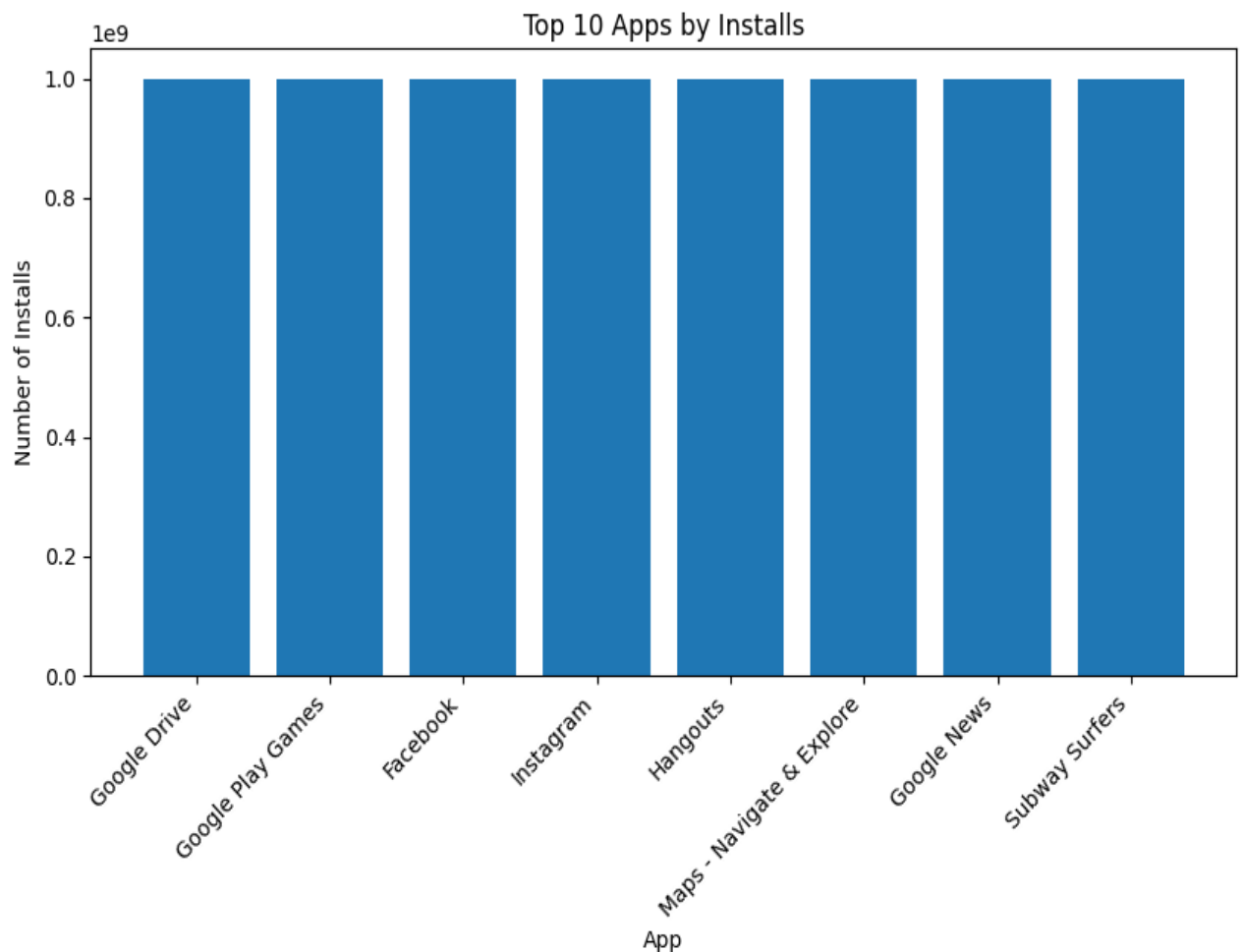
1. TOP 10 LARGEST APPLICATIONS:

A BAR CHART WAS CREATED TO IDENTIFY APPLICATIONS WITH THE LARGEST FILE SIZES.



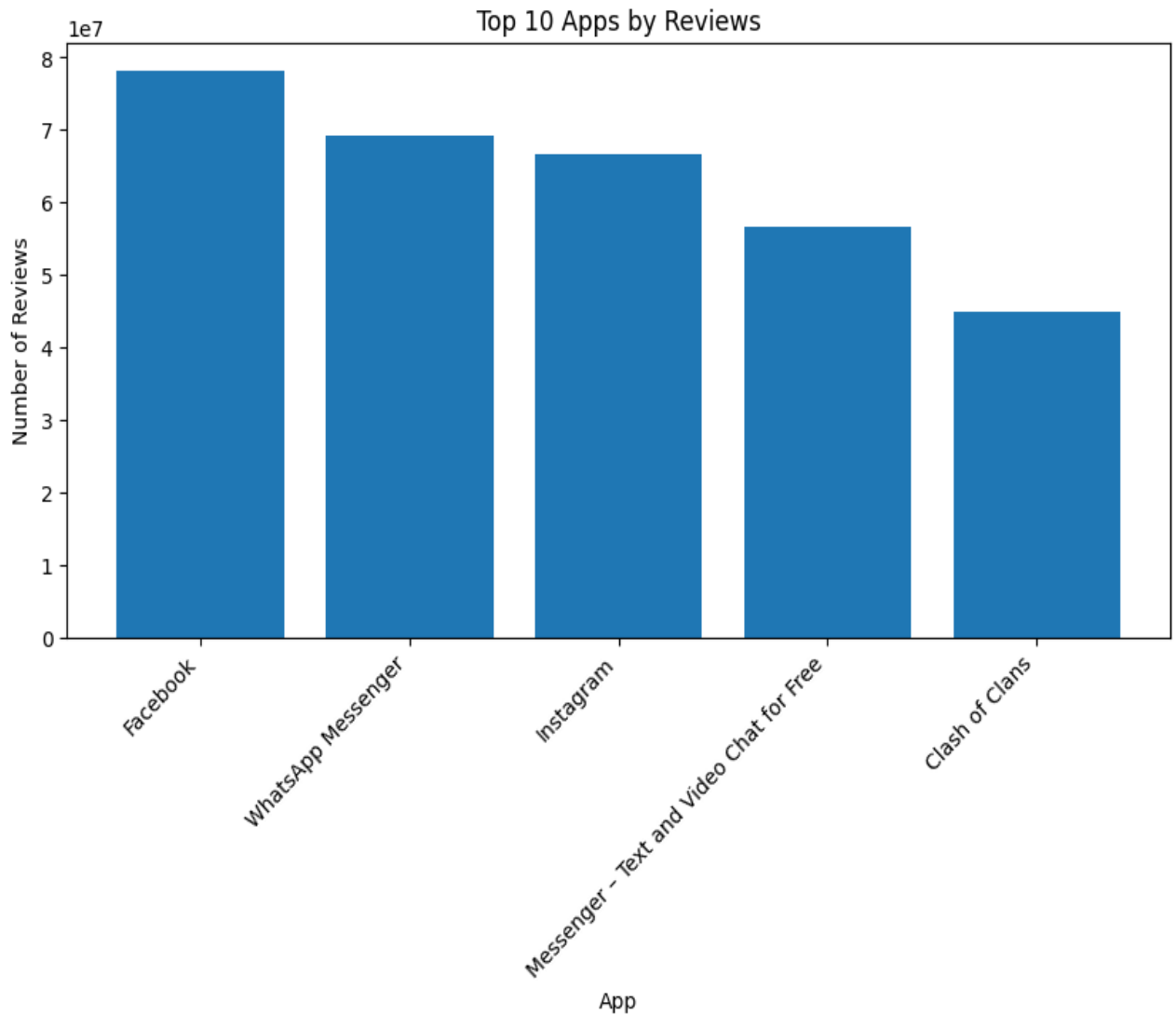
2. TOP INSTALLED APPLICATIONS:

A BAR CHART WAS USED TO IDENTIFY APPLICATIONS WITH THE HIGHEST NUMBER OF INSTALLATIONS.



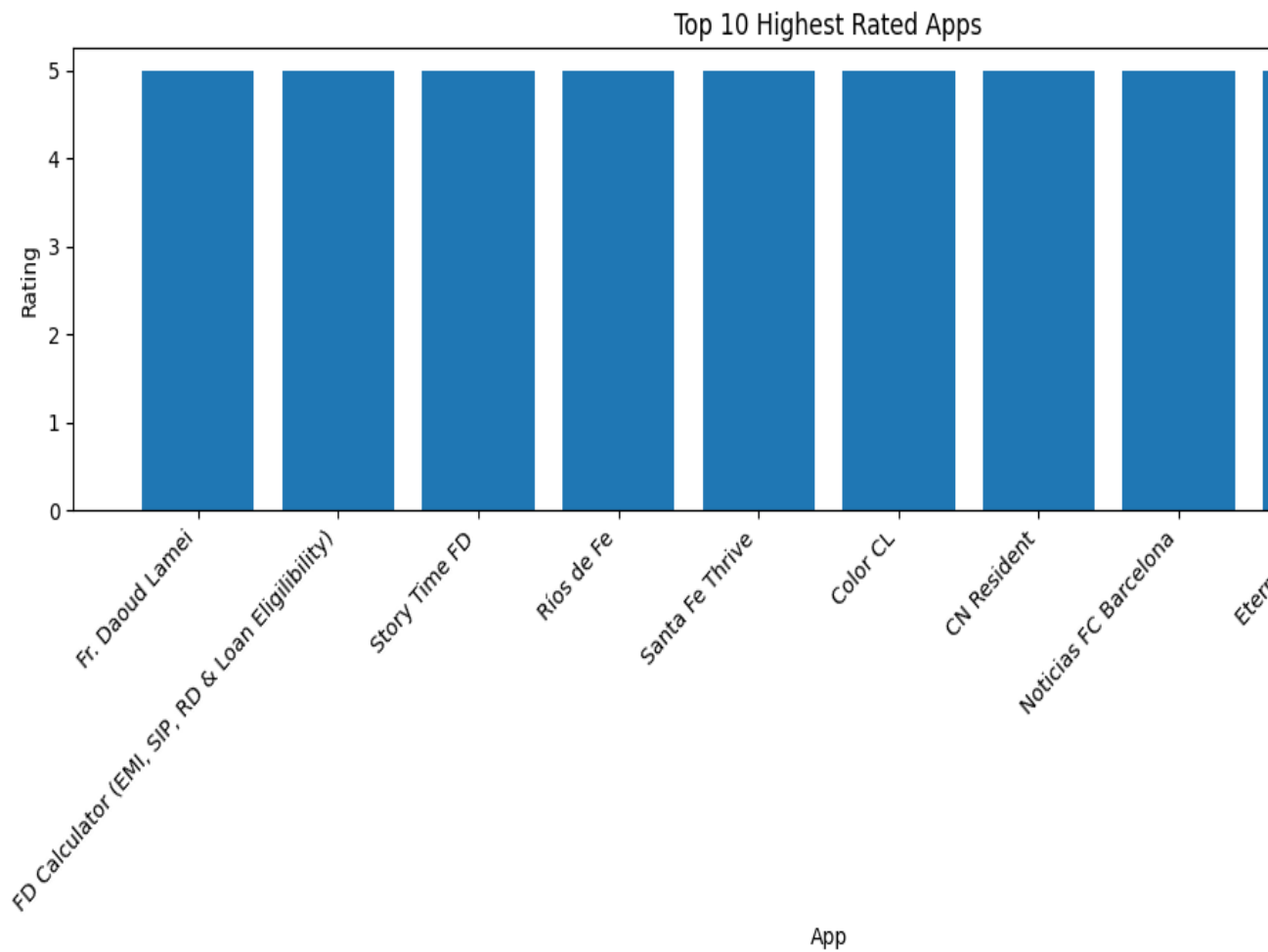
3. TOP REVIEWED APPLICATIONS:

A BAR CHART WAS CREATED TO IDENTIFY APPLICATIONS WITH THE HIGHEST NUMBER OF USER REVIEWS.



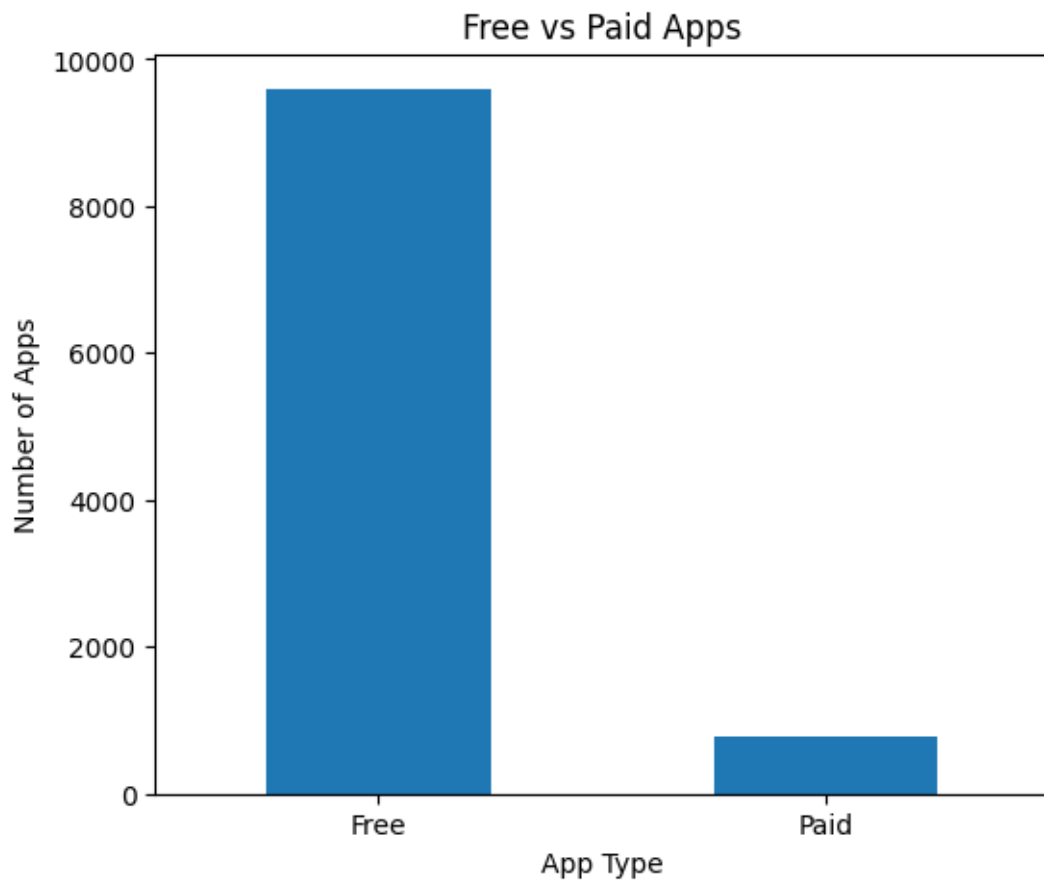
4. APP RATING ANALYSIS:

A VISUALIZATION WAS CREATED TO ANALYZED HIGHLY RATED APPLICATIONS AND UNDERSTAND USER RATINGS.



5. FREE AND PAID APPLICATIONS:

THE APPLICATIONS WERE ANALYZED BASED ON THEIR TYPE TO UNDERSTAND THE DIFFERENCE BETWEEN FREE AND PAID APPS.



THESE VISUALIZATIONS MAKE IT EASIER TO IDENTIFY PATTERNS, COMPARE APPLICATIONS, AND UNDERSTAND THE RESULTS OF THE ANALYSIS.

8. KEY INSIGHTS

COPY AND PASTE THIS INTO YOUR WORD DOCUMENT:

8. KEY INSIGHTS

THE ANALYSIS OF THE GOOGLE PLAY STORE DATASET PROVIDED THE FOLLOWING IMPORTANT INSIGHTS:

FREE APPS HAVE HIGHER USER ADOPTION:

FREE APPLICATIONS GENERALLY HAVE MUCH HIGHER INSTALLATION NUMBERS COMPARED TO PAID APPLICATIONS.

POPULAR APPS HAVE HIGH INSTALLATIONS:

APPLICATIONS WITH HIGH POPULARITY USUALLY HAVE A VERY LARGE NUMBER OF INSTALLATIONS.

SOCIAL AND COMMUNICATION APPS RECEIVE MANY REVIEWS:

POPULAR SOCIAL AND COMMUNICATION APPLICATIONS RECEIVE A LARGE NUMBER OF USER REVIEWS, SHOWING HIGH USER ENGAGEMENT.

SOME APPS HAVE LARGE FILE SIZES:

THE ANALYSIS IDENTIFIED APPLICATIONS WITH VERY LARGE FILE SIZES, WHICH MAY REQUIRE MORE STORAGE SPACE ON USERS' DEVICES.

HIGHLY RATED APPS SHOW POSITIVE USER FEEDBACK:

HIGHLY RATED APPLICATIONS INDICATE POSITIVE USER FEEDBACK AND SATISFACTION.

INSTALLS AND REVIEWS REFLECT USER ENGAGEMENT:

THE NUMBER OF INSTALLATIONS AND REVIEWS PROVIDES USEFUL INFORMATION ABOUT THE POPULARITY AND ENGAGEMENT OF APPLICATIONS.

OVERALL INSIGHT

OVERALL, THE ANALYSIS SHOWS THAT INSTALLATIONS, REVIEWS, RATINGS, PRICING, AND APP SIZE ARE IMPORTANT CHARACTERISTICS FOR UNDERSTANDING GOOGLE PLAY STORE APPLICATIONS AND USER ENGAGEMENT.

9. FINAL CONCLUSION

THIS PROJECT ANALYZED GOOGLE PLAY STORE APP DATA USING PYTHON TO UNDERSTAND APP POPULARITY, USER ENGAGEMENT, RATINGS, PRICING, AND STORAGE REQUIREMENTS.

THE ANALYSIS SHOWED THAT FREE APPLICATIONS GENERALLY ACHIEVE HIGHER INSTALLATION NUMBERS, WHILE POPULAR SOCIAL AND COMMUNICATION APPLICATIONS RECEIVE A LARGE NUMBER OF USER REVIEWS. THE PROJECT ALSO IDENTIFIED HIGHLY RATED APPLICATIONS AND APPLICATIONS WITH LARGE FILE SIZES.

OVERALL, THE PROJECT HELPED IN UNDERSTANDING HOW DATA ANALYSIS CAN BE USED TO IDENTIFY PATTERNS AND GENERATE USEFUL INSIGHTS FROM REAL-WORLD DATASETS. THE USE OF PYTHON, PANDAS, NUMPY, MATPLOTLIB, AND SEABORN MADE IT POSSIBLE TO CLEAN, ANALYZE, AND VISUALIZE THE GOOGLE PLAY STORE DATA EFFECTIVELY.

10. FUTURE SCOPE

THE GOOGLE PLAY STORE DATA ANALYSIS PROJECT CAN BE FURTHER IMPROVED AND EXPANDED IN THE FUTURE. SOME POSSIBLE AREAS FOR FUTURE ANALYSIS ARE:

CATEGORY-WISE ANALYSIS:

ANALYZE DIFFERENT APP CATEGORIES TO IDENTIFY WHICH CATEGORIES HAVE HIGHER INSTALLATIONS, RATINGS, AND REVIEWS.

CORRELATION ANALYSIS:

STUDY THE RELATIONSHIP BETWEEN RATINGS, REVIEWS, INSTALLATIONS, APP SIZE, AND PRICE.

TIME-BASED ANALYSIS:

ANALYZE APP UPDATES AND CHANGES OVER TIME TO UNDERSTAND HOW APP PERFORMANCE CHANGES.

PREDICTIVE ANALYSIS:

MACHINE LEARNING MODELS CAN BE USED TO PREDICT APP POPULARITY OR INSTALLATION LEVELS.

SENTIMENT ANALYSIS:

IF REVIEW TEXT DATA IS AVAILABLE, SENTIMENT ANALYSIS CAN BE PERFORMED TO UNDERSTAND WHETHER USER FEEDBACK IS POSITIVE, NEGATIVE, OR NEUTRAL.

INTERACTIVE DASHBOARD:

THE ANALYSIS CAN BE CONVERTED INTO AN INTERACTIVE DASHBOARD USING POWER BI OR OTHER DATA VISUALIZATION TOOLS.

CONCLUSION OF FUTURE SCOPE

THESE IMPROVEMENTS CAN PROVIDE DEEPER INSIGHTS INTO APPLICATION PERFORMANCE, USER PREFERENCES, AND MARKET TRENDS. THEY CAN ALSO HELP DEVELOPERS AND BUSINESSES MAKE BETTER DATA-DRIVEN DECISIONS.